

IC 101P: Reverse Engineering

Course Code	IC 101P
Course Name	Reverse Engineering
L-T-P-C	0-0-3-2
Prerequisites	Consent of the faculty member
Intended for	B.Tech
Distribution	Core
Approval	2nd Adhoc Senate (May 2012), in the B.Tech Institute Core (common core courses) curriculum
Revised	47th BoA (Sep 2022) - renamed from “Reverse Engineering”

Course Description

The students will focus on either software or hardware reverse engineering (RE). In the process of RE students understand existing technologies, functions, features, objects, components and systems. By carefully disassembling, observing, testing, analyzing and reporting, students can understand how something works and suggest ways it might be improved.

This process requires careful observation, disassembly, documentation, analysis and reporting. Many times, the reverse engineering process is non-destructive. This means that the object or component can be reassembled and still function just as it did before it was taken apart.

Throughout the reverse engineering project, the students are able to think of ways these objects could be improved. Is there some way it could function better? or manufactured less expensively? The students will use observations to make suggestions for improvement of the product.

Course Objectives

After the completion of this course, students should be able to:

- The basic understanding of engineering systems.
- Understanding the terminologies related to re-engineering, forward engineering, and reverse engineering.
- Disassemble products and specify the interactions between its subsystems and their functionality
- The Understanding of Reverse Engineering methodologies.
- Understanding of Reverse engineering of Systems, Mechanical RE, Electronic RE, and Computer RE.

Key Learning Topics

Forward Engineering Design, Design Thought and Process, Design Steps, System RE, RE Methodology, RE Steps, System level Design, and Examples, Product Development, Product Functions, Engineering Specifications, Product Architecture, Mechanical RE, Computer-Aided RE, Electronic RE, Identify electronic components, PCB RE, Schematic Drawings and Analysis, S/W RE, Reverse Engineering in Computer Applications, Re-engineering of PLC programs.

References

1. *Product Design: Techniques in Reverse Engineering and New Product Development* by K. Otto and K. Wood, Prentice Hall, 2001.
2. *Reverse Engineering: An Industrial Perspective* by Raja and Fernandes. Springer-Verlag 2008
3. RE as necessary phase by rapid product development by Sokovic and Kopac. *Journal of Materials Processing Technology* 2005
4. *Reversing: Secrets of Reverse Engineering* by Eldad Eilam Publisher: Wiley (April 15, 2005)
5. *The IDA Pro Book: The Unofficial Guide to the World's Most Popular Disassembler* by Chris Eagle